

MOBILE APPLICATION DEVELOPMENT(THEORY)

ASSIGNMENT # 1

SUBMITTED BY:

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Batch File Fundamentals

What is a Batch File?

A **Batch file** is a text file containing one or more commands that are executed sequentially by the Windows command interpreter.

Typical extensions are .bat and .cmd. For this lab, we mainly use **.bat**.

Why Do We Use Batch Scripts?

Batch scripts can be used for:

- Creating folders automatically
- Copying files
- Creating backups
- Renaming files
- Deleting temporary files
- Running programs
- Checking conditions
- Repeating commands
- Displaying messages
- Performing simple administration tasks

Real-World Example

Suppose an employee has to copy all .txt files into a Backup folder every evening.

Manual method	Batch method
Create Backup folder	One script performs the complete task
Find files	
Copy files	
Check result	

Core Script Elements

PRACTICAL1.bat

File Edit View

```
@echo off
mkdir Backup
copy *.txt Backup
echo Backup completed.
pause
```

OUTPUT:

C:\Windows\system32\cmd.e

```
A subdirectory or file Backup already exists.
PRACTICAL1.txt
        1 file(s) copied.
Backup completed.
Press any key to continue . . .
```

@echo off

Most Batch scripts start with @echo off. It prevents each command itself from being displayed before its output.

Without @echo off

PRACTICAL1.bat practical2.bat

File Edit View

```
echo Hello Students
pause
```

OUTPUT:

C:\Windows\system32\cmd.e

```
C:\Users\123\Documents>echo Hello Students
Hello Students

C:\Users\123\Documents>pause
Press any key to continue . . .
```

With @echo off:

PRACTICAL1.bat | practical2.bat | practical3.bat

File Edit View

```
@echo off
echo Hello
pause
```

Comments in Batch Files

Comments help programmers explain what the script does. A common method is:

File Edit View

```
@echo off

REM Display welcome message
echo Welcome to OS Lab

pause
```

OUTPUT:

C:\Windows\system32\cmd.e

```
Welcome to OS Lab
Press any key to continue . . .
```

Basic Batch Script Structure

File Edit View

```
@echo off

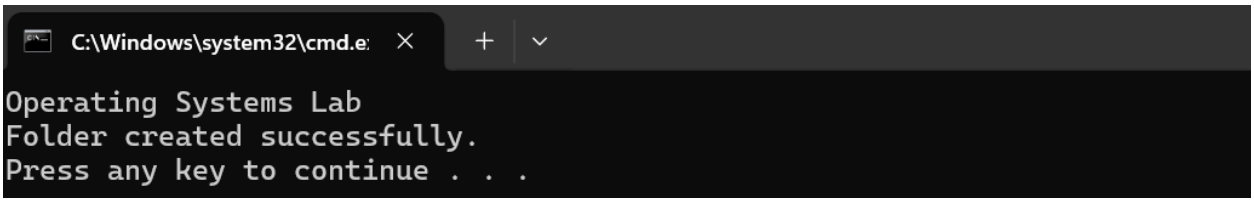
REM Display heading
echo Operating Systems Lab

REM Create folder
mkdir OS_Lab

REM Display completion message
echo Folder created successfully.

pause
```

OUTPUT:



Creating Folders

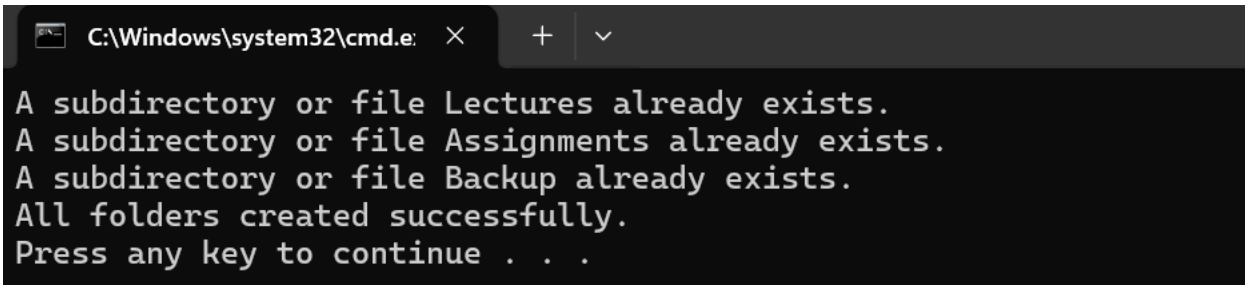
Practical 1 — Automatically Create Folders

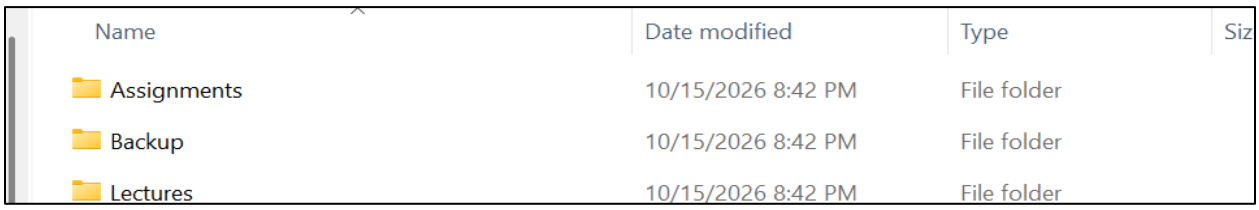
Objective: Create multiple directories automatically using one Batch file.

File Edit View

```
@echo off
REM Create main folders
mkdir Lectures
mkdir Assignments
mkdir Backup
echo All folders created successfully.
pause
```

OUTPUT:





Practical 2 — Create Folder Structure

File Edit View

```
@echo off

REM Create main directory
mkdir Operating_Systems

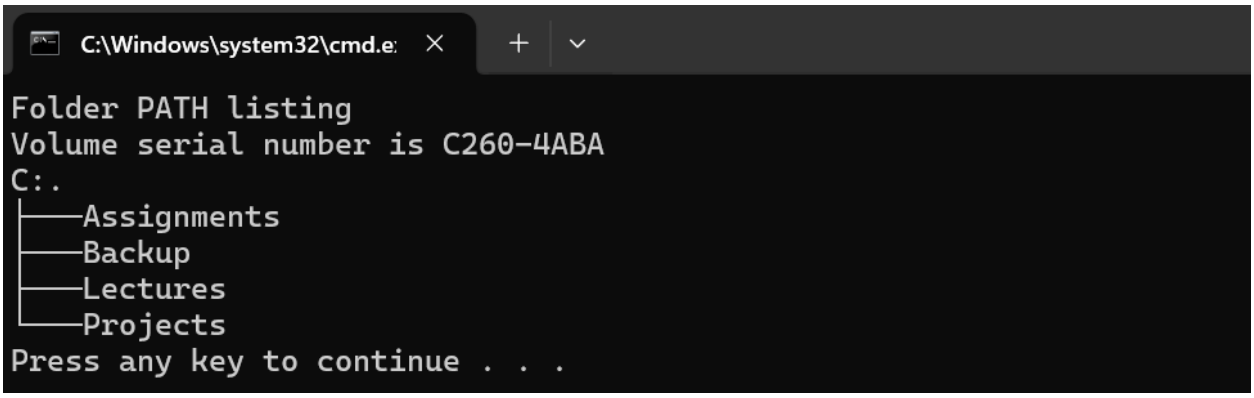
REM Enter main directory
cd Operating_Systems

REM Create subdirectories
mkdir Lectures
mkdir Assignments
mkdir Projects
mkdir Backup

REM Display structure
tree

pause
```

OUTPUT:



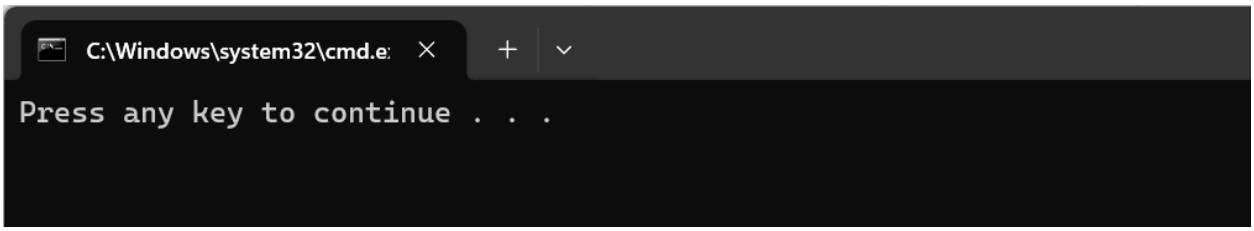
Creating Files and Output Redirection

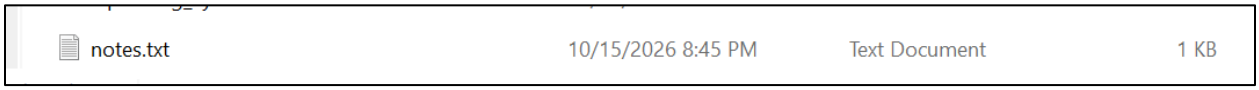
Creating Files Through a Batch Script

```
File    Edit    View

@echo off
echo Operating Systems Lab > notes.txt
pause
```

OUTPUT:





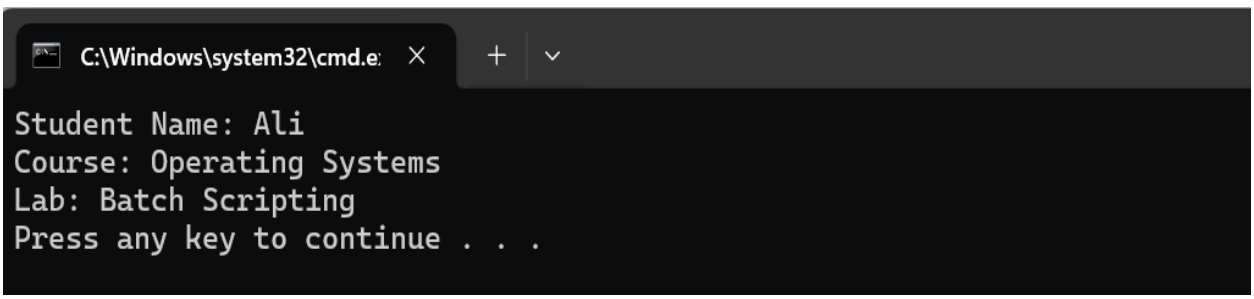
Practical 3 — Create a Student Information File

```
File    Edit    View

@echo off
echo Student Name: Ali > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt
type student.txt

pause
```

OUTPUT:

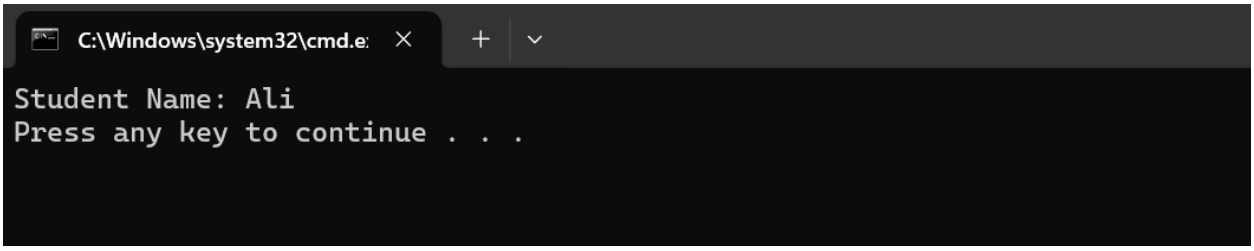


Variables and User Input

Variables

```
@echo off
REM Store student's name
set name=Ali
REM Display variable
echo Student Name: %name%
pause
```

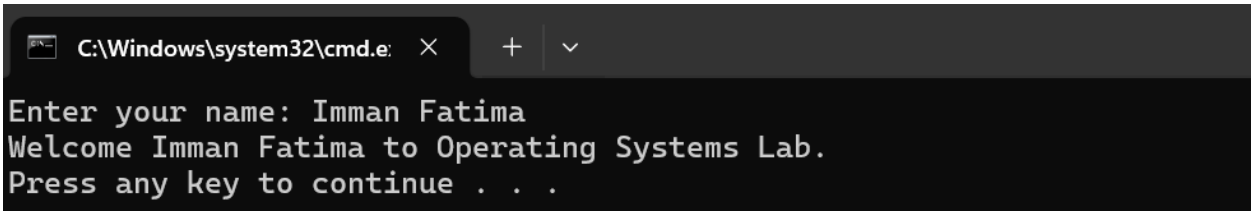
OUTPUT:



Practical 4 — User Input

```
@echo off
REM Ask user for name
set /p name=Enter your name:
REM Display personalized message
echo Welcome %name% to Operating Systems Lab.
pause
```

OUTPUT:



Multiple Inputs

```
File    Edit    View

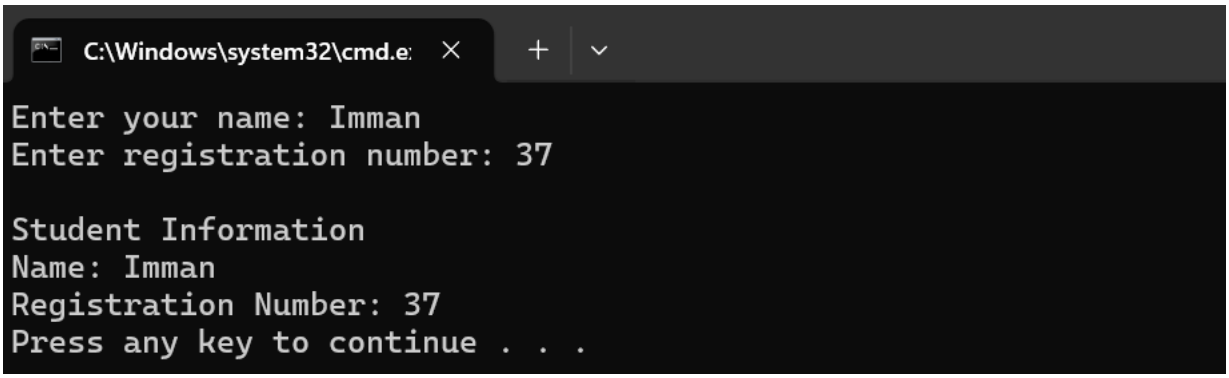
@echo off

set /p name=Enter your name:
set /p reg=Enter registration number:

echo.
echo Student Information
echo Name: %name%
echo Registration Number: %reg%

pause
```

OUTPUT:



Arithmetic in Batch

Batch can perform integer arithmetic using set /a.

Practical 5 — Simple Addition

```
File    Edit    View

@echo off

REM Take first number
set /p num1=Enter first number:

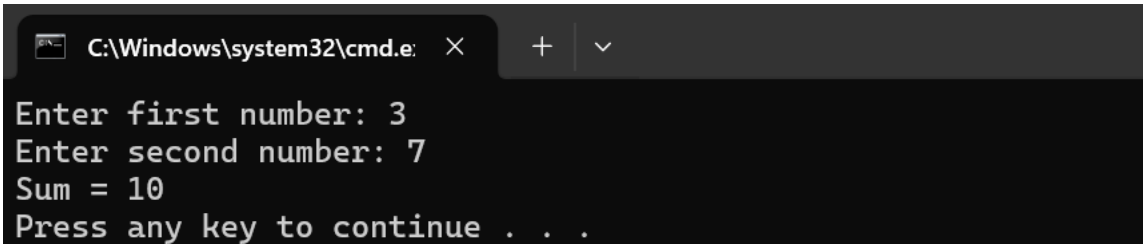
REM Take second number
set /p num2=Enter second number:

REM Add numbers
set /a result=num1+num2

REM Display result
echo Sum = %result%

pause
```

OUTPUT:



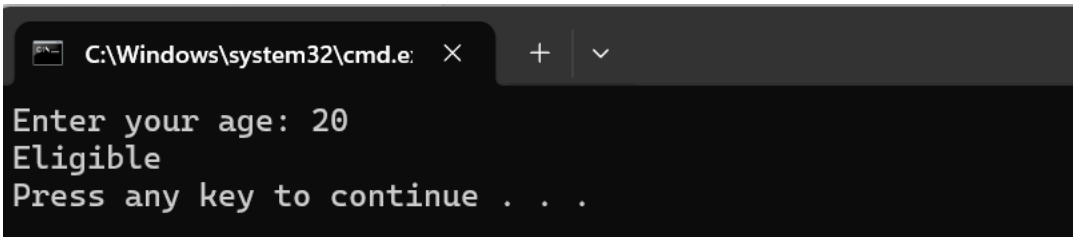
Conditions: IF / ELSE

File Edit View

```
@echo off
set /p age=Enter your age:

if %age% GEQ 18 (
    echo Eligible
) else (
    echo Not Eligible
)
pause
```

OUTPUT:



Practical 6 — Pass/Fail Script

File Edit View

```
@echo off

REM Ask student marks
set /p marks=Enter marks:

REM Check result
if %marks% GEQ 50 (
    echo Result: PASS
) else (
    echo Result: FAIL
)

pause
```

OUTPUT:



File Existence Checks

Backup Only If File Exists

```
@echo off

REM Check if file exists
if exist report.txt (

    REM Create backup directory if needed
    if not exist Backup mkdir Backup

    REM Copy file
    copy report.txt Backup

    echo Backup completed.

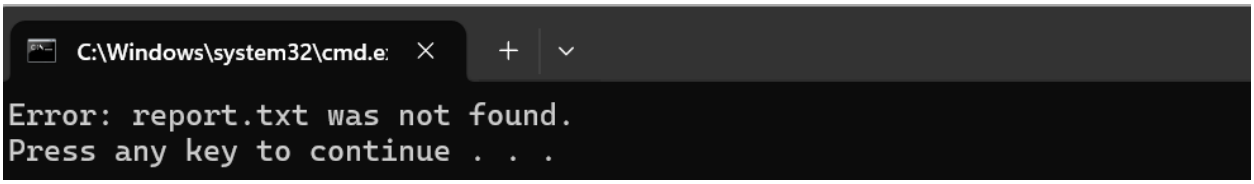
) else (

    echo Error: report.txt was not found.

)

pause
```

OUTPUT:



Loops: FOR

A loop repeats commands automatically. Batch commonly uses for.

Practical 7 — Create Multiple Folders Using a Loop

File Edit View

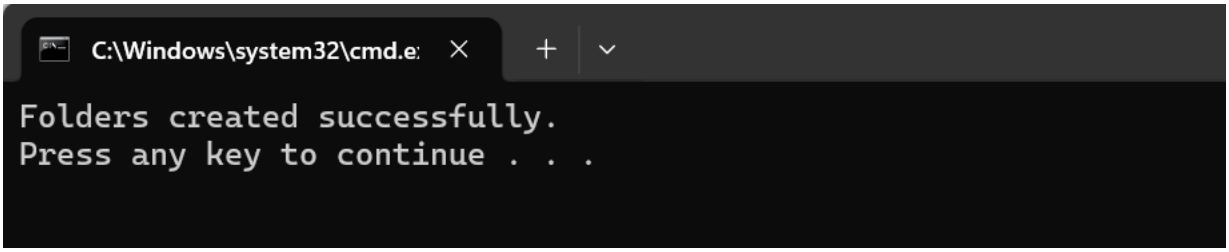
```
@echo off

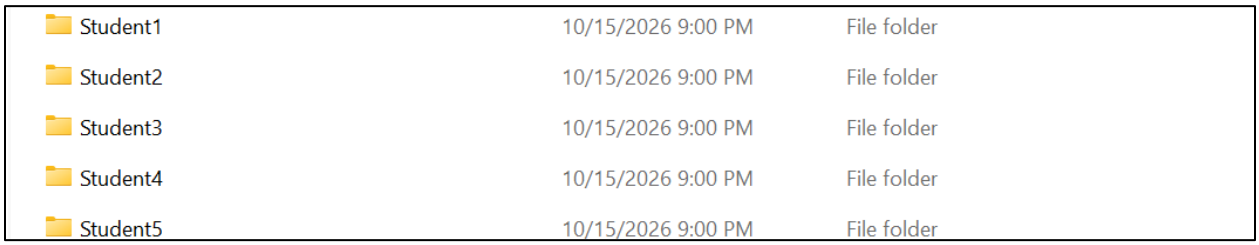
for %%i in (1 2 3 4 5) do (
    mkdir Student%%i
)

echo Folders created successfully.

pause
```

OUTPUT:





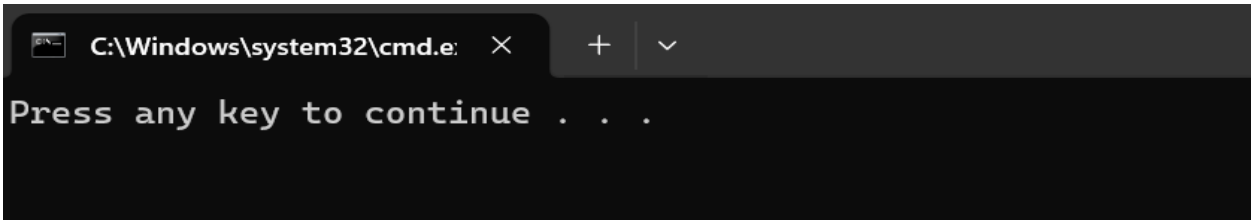
Create 10 Folders Automatically

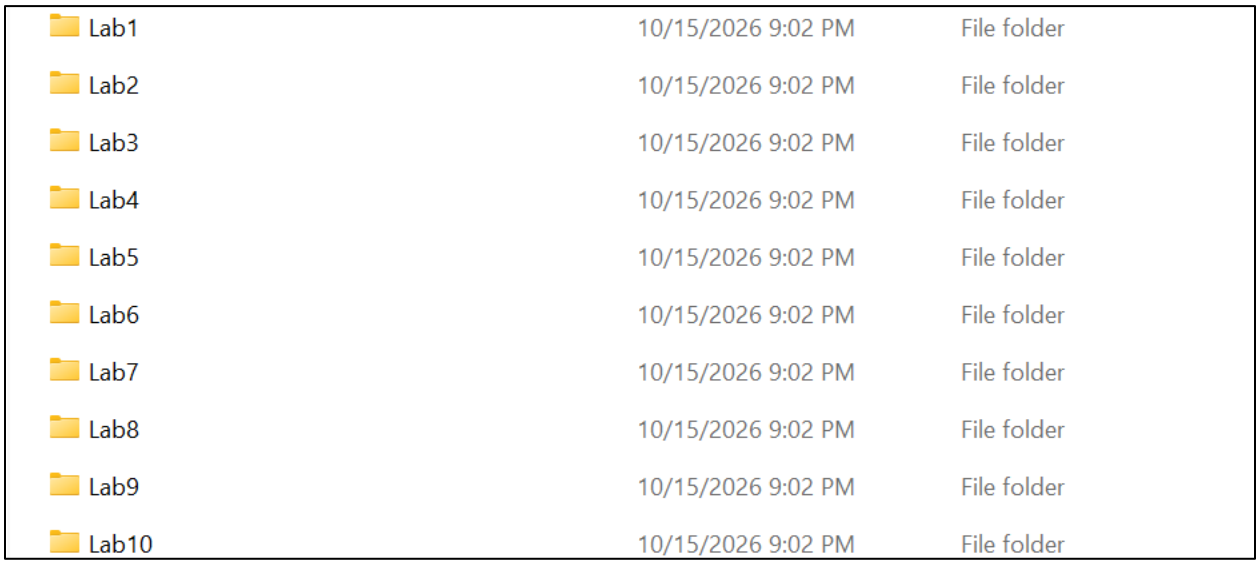
```
@echo off

for /L %%i in (1,1,10) do (
    mkdir Lab%%i
)

pause
```

OUTPUT:





Practical 8 — Complete Batch Automation

Objective: Create a script that automatically:

- Displays a title
- Accepts student name
- Creates a folder
- Creates a text file
- Stores student information
- Creates Backup directory
- Copies file
- Displays completion message

```
@echo off

REM Set CMD window title
title OS Lab Student Automation

REM Ask for student name
set /p name=Enter student name:

REM Create main folder
mkdir Student_Record

REM Enter folder
cd Student_Record

REM Create student information file
echo Student Name: %name% > student.txt
echo Course: Operating Systems >> student.txt
echo Lab: Batch Scripting >> student.txt

REM Create backup folder if it does not exist
if not exist Backup mkdir Backup

REM Copy information file into backup
copy student.txt Backup

REM Display contents
echo.
echo Student Record:
type student.txt
echo.
echo Automation completed successfully.
pause
```

OUTPUT

